

Multivariate Gaussian Density Functions, Covariance Matrices, and Constant Density Curves

Lab Assignment 4 Report

Shrawanthi P. (Roll No: 26120098)
Shreepriyadarshini R.M. (Roll No: 26120099)

19th September 2026

1 Introduction and Objectives

In this assignment, we analyze how different structures of the covariance matrix affect the shape, spread, and orientation of data distributions in a 2-D gaussian random variables. We also explore how decision boundaries are formed between two Gaussian classes under different covariance conditions.

The primary objectives are:

1. Generate 2-D Gaussian data under three types of covariance matrices:
 - **Isotropic Covariance Matrix:** $C = \sigma^2 I$
 - **Diagonal Covariance Matrix:** $C = \begin{bmatrix} \sigma_{11}^2 & 0 \\ 0 & \sigma_{22}^2 \end{bmatrix}$
 - **Full Covariance Matrix:** $C = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{21} & \sigma_{22}^2 \end{bmatrix}$
2. Transform standard Gaussian samples using the matrix square root: $Y = \mu + \Sigma^{1/2}(X - \mu)$.
3. Estimate the sample mean and the covariance matrix from the generated data and then compare them with the target parameters.
4. Compute the eigenvalues and eigenvectors, and then plot the constant density curves (Mahalanobis distance contours) along with the principal axes.
5. Investigate the Bayes decision boundaries between two Gaussian classes across three different covariance configurations (linear vs. quadratic boundaries).

2 Formulae and Methodology

2.1 Multivariate Gaussian Density

A 2-D Gaussian vector $x = [x_1, x_2]^T$ with the mean vector $\mu \in \mathbb{R}^2$ and covariance matrix $\Sigma \in \mathbb{R}^{2 \times 2}$ is given by:

$$p(x) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right) \quad (1)$$

2.2 Constant Density Curves and Mahalanobis Distance

Curves of constant probability density correspond to points with a constant Mahalanobis distance:

$$d_M^2(x, \mu) = (x - \mu)^T \Sigma^{-1} (x - \mu) = c^2 \quad (2)$$

In our experiments, we plot contours for $c^2 \in \{1, 4, 9\}$, which correspond to 1, 2, and 3 standard deviations from the mean ($1\sigma, 2\sigma, 3\sigma$).

- The directions of the principal axes of the ellipses are given by the **eigenvectors** v_1, v_2 of Σ .
- The lengths of the semi-axes are proportional to the square root of the **eigenvalues**, $c\sqrt{\lambda_i}$.

2.3 Data Generation via Matrix Square Root

Starting from uncorrelated Gaussian variables $X \sim \mathcal{N}(0, I)$, we generate samples $Y \sim \mathcal{N}(\mu, \Sigma)$ using:

$$Y = \mu + \Sigma^{1/2}(X - \mu) \quad (3)$$

To compute any symmetric positive semi-definite covariance matrix Σ , the matrix square root, use:

$$\Sigma = V\Lambda V^T \implies \Sigma^{1/2} = V\Lambda^{1/2}V^T \quad (4)$$

where $V = [v_1, v_2]$ is the matrix of eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \lambda_2)$ contains the eigenvalues.

2.4 Bayes Decision Boundaries between Two Classes

For two equiprobable Gaussian classes $\omega_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$ and $\omega_2 \sim \mathcal{N}(\mu_2, \Sigma_2)$, the decision boundary satisfies the following $p(x|\omega_1) = p(x|\omega_2)$:

$$(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) + \ln |\Sigma_1| = (x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) + \ln |\Sigma_2| \quad (5)$$

- **Equal Covariance** ($\Sigma_1 = \Sigma_2 = \Sigma$): Here the quadratic terms cancel out. The decision boundary is **linear**:

$$w^T x + w_0 = 0, \quad \text{where } w = \Sigma^{-1}(\mu_1 - \mu_2) \quad (6)$$

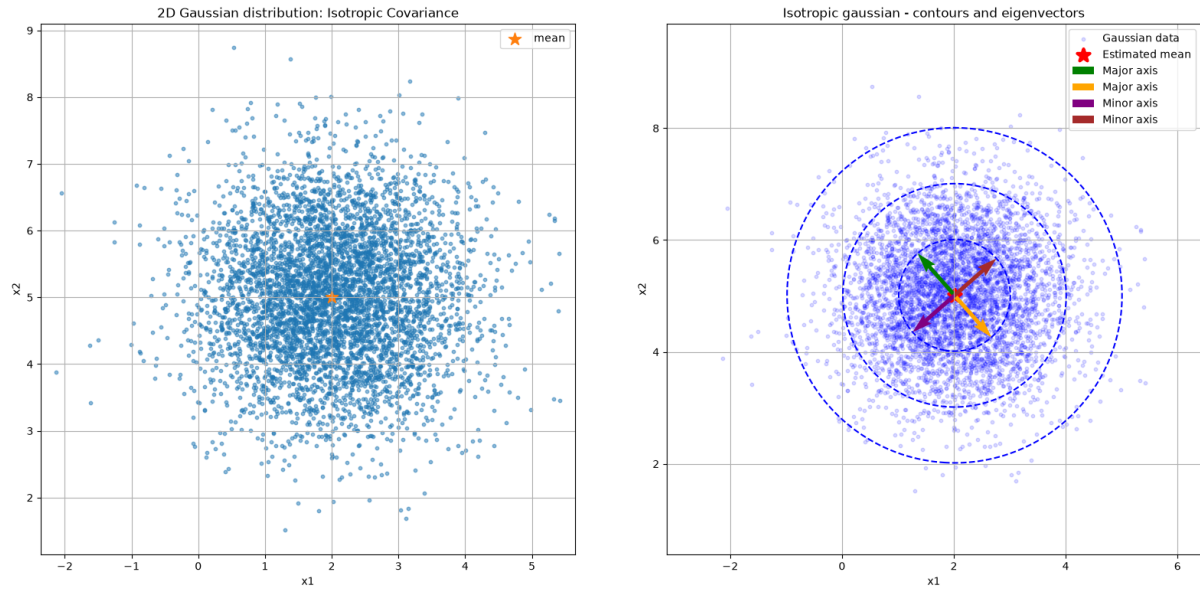
- **Unequal Covariance** ($\Sigma_1 \neq \Sigma_2$): The quadratic terms do not cancel, thus produces a **quadratic decision boundary** (hyperbolic, elliptical, or parabolic).

3 Experiments and Results

3.1 Problem 1: Isotropic Covariance Matrix ($C = \sigma^2 I$)

We generated $n = 5000$ points with mean $\mu = [2.0, 5.0]^T$ and variance $\sigma^2 = 1.0$.

- **Target Parameters:** $\mu = [2.0, 5.0]^T$, $\Sigma = \begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$.
- **Estimated Parameters:** $\hat{\mu} = [2.00, 4.99]^T$, $\hat{\Sigma} = \begin{bmatrix} 0.99 & 0.01 \\ 0.01 & 1.01 \end{bmatrix}$.
- **Eigenvalues:** $\lambda_1 \approx 1.01$, $\lambda_2 \approx 0.99$.



(a) Raw generated data points with mean marked. (b) Circular contours ($1\sigma, 2\sigma, 3\sigma$) and eigenvectors.

Figure 1: 2-D Gaussian distribution with isotropic covariance matrix $C = \sigma^2 I$.

Observations:

- In the above Figure 1, the constant density curves are **concentric circles**.
- The variances along both axes are equal ($\sigma_1^2 = \sigma_2^2 = 1$), and the cross-covariance is zero.
- The eigenvalues are approximately same ($\lambda_1 \approx \lambda_2 \approx 1$), so that the spread is uniform in all directions around the contour.
- Both major axis and minor axis are of the same length and point in directions that is not aligned to the x1 and x2 axes.

3.2 Problem 2: Diagonal Covariance Matrix

We used target mean $\mu = [2.0, 5.0]^T$ and diagonal covariance matrix:

$$\Sigma = \begin{bmatrix} 4.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}, \quad \Sigma^{1/2} = \begin{bmatrix} 2.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$$

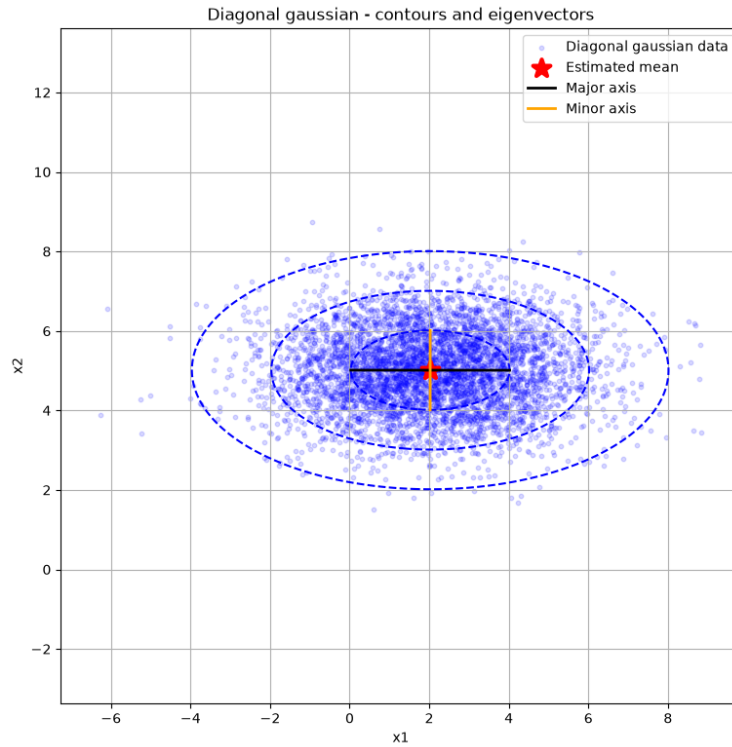


Figure 2: 2-D Gaussian distribution with diagonal covariance matrix showing axis-aligned elliptical contours and principal axes.

Observations:

- In Figure 2, the constant density curves are ellipses aligned with the coordinate axes x_1 and x_2 .
- The major axis lies along x_1 with length proportional to $\sqrt{\lambda_1} = \sqrt{4} = 2.0$, and the minor axis lies along x_2 with length proportional to $\sqrt{\lambda_2} = \sqrt{1} = 1.0$.
- The eigenvectors are exactly along the basis vectors $[1, 0]^T$ and $[0, 1]^T$.

3.3 Problem 3: Full Covariance Matrix

We now try a full covariance matrix with positive cross-correlation:

$$\Sigma = \begin{bmatrix} 4.0 & 1.5 \\ 1.5 & 1.0 \end{bmatrix}$$

Table 1 compares the target and estimated statistics.

Table 1: Comparison of Target and Sample Statistics for Full Covariance Matrix

Parameter	Target (Theoretical)	Estimated from Data
Mean Vector μ	$[2.0, 5.0]^T$	$[2.01, 5.01]^T$
Covariance Matrix Σ	$\begin{bmatrix} 4.0 & 1.5 \\ 1.5 & 1.0 \end{bmatrix}$	$\begin{bmatrix} 3.98 & 1.49 \\ 1.49 & 1.01 \end{bmatrix}$
Eigenvalues (λ_1, λ_2)	(4.62, 0.38)	(4.60, 0.38)

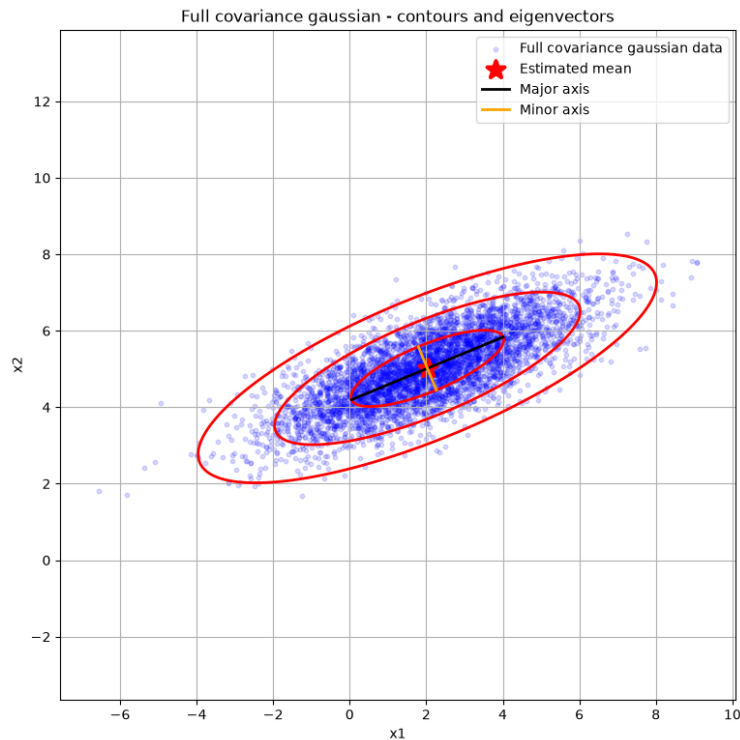


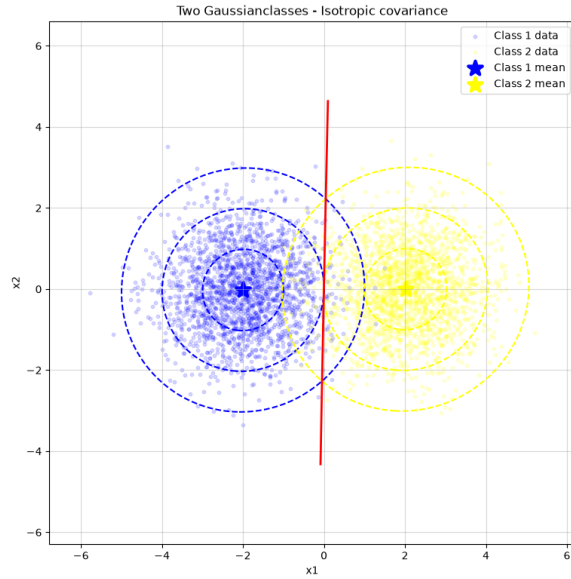
Figure 3: 2-D Gaussian distribution with full covariance matrix showing rotated elliptical contours and tilted eigenvectors.

Observations:

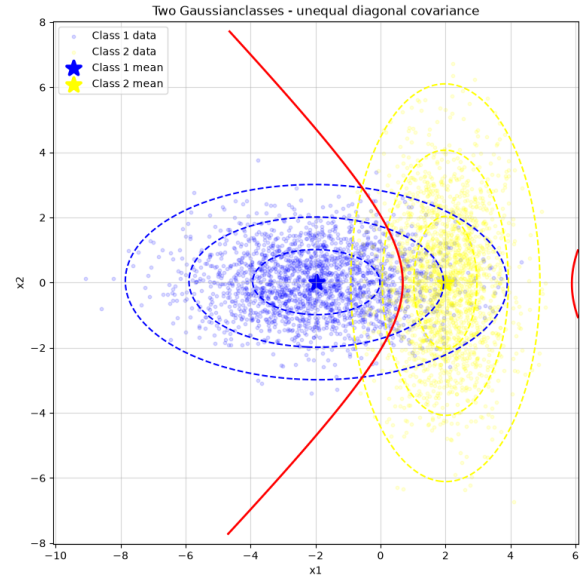
- As shown in the above Figure 3, the constant density curves are tilted ellipses.
- The positive covariance ($\sigma_{12} = 1.5$) causes the distribution to tilt diagonally upwards.
- Here the major axis aligns with the eigenvector of the largest eigenvalue ($\lambda_1 = 4.62$), while the minor axis corresponds to $\lambda_2 = 0.38$.

3.4 Problem 4: Decision Boundaries between Two Classes

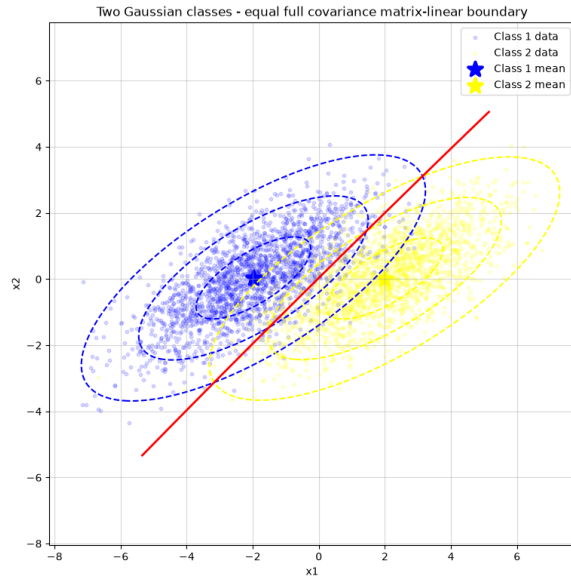
We then tried all three cases for 2 Gaussian classes $\omega_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$ and $\omega_2 \sim \mathcal{N}(\mu_2, \Sigma_2)$ with $n = 2000$ samples each. We observed these four visual plots for all cases as presented in Figure 4.



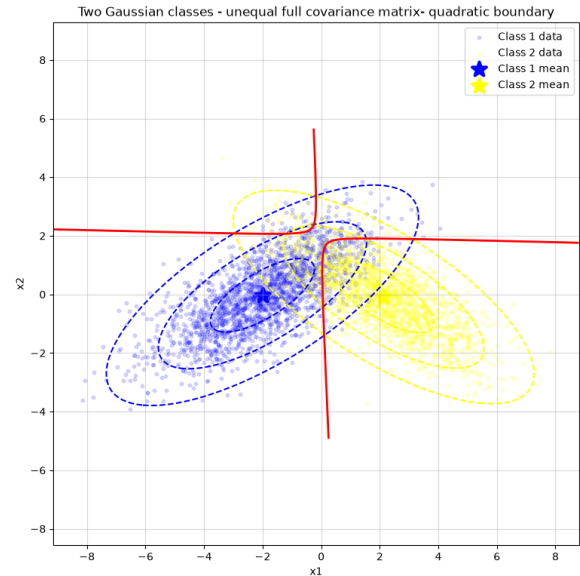
(a) Case 1: Equal isotropic ($\Sigma_1 = \Sigma_2 = I$) $\rightarrow$ Linear.



(b) Case 2: Unequal diagonal $\rightarrow$ Quadratic.



(c) Case 3: Equal full covariance $\rightarrow$ Linear.



(d) Case 4: Unequal full covariance $\rightarrow$ Quadratic.

Figure 4: Bayes decision boundaries (solid red line) between two Gaussian classes under four covariance conditions.

3.4.1 Detailed Analysis of Boundary Cases

1. **Case 1: Equal Isotropic Covariances** ($\Sigma_1 = \Sigma_2 = I$): Means are $\mu_1 = [-2, 0]^T$ and $\mu_2 = [2, 0]^T$. Since the quadratic term cancels out, the decision boundary is a straight vertical line ($x_1 = 0$), which acts as the perpendicular bisector between the 2 class means.
2. **Case 2: Unequal Diagonal Covariances** ($\Sigma_1 = \text{diag}(4, 1)$, $\Sigma_2 = \text{diag}(1, 4)$): Since the variances are unequal along the axes, the quadratic terms do not cancel which is why the decision boundary is curved (quadratic hyperbola), bending around the class with smaller variance.

3. **Case 3: Equal Full Covariances** ($\Sigma_1 = \Sigma_2 = \Sigma$): Both classes share $\Sigma = \begin{bmatrix} 3.0 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}$. Because the covariances are equal, so the boundary is strictly linear. It is tilted at an angle dictated by $\Sigma^{-1}(\mu_1 - \mu_2)$.
4. **Case 4: Unequal Full Covariances** ($\Sigma_1 \neq \Sigma_2$): Class 1 has positive correlation ($\sigma_{12} = +1.5$) while Class 2 has negative correlation ($\sigma_{12} = -1.5$). The opposite orientations produces two hyperbolic branches of a quadratic boundary.

4 Discussion and Key Findings

- **Role of Covariance Structure:**

- *Isotropic* ($\Sigma = \sigma^2 I$): This produces circular contours and the variance is identical in all directions.
- *Diagonal* ($\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$): This produces axis-aligned elliptical contours.
- *Full Covariance*: This produces rotated elliptical contours where the angle of rotation corresponds to the principal eigenvector.

- **Eigenvalues and Eigenvectors:** The eigenvectors define the directions of the principal axes, and the eigenvalues quantify the variance/spread along each axis (length $\propto \sqrt{\lambda_i}$).

- **Nature of Decision Boundaries:**

- When covariance matrices are identical ($\Sigma_1 = \Sigma_2$), the boundary is always linear.
- When covariance matrices differ ($\Sigma_1 \neq \Sigma_2$), the boundary is always quadratic.

5 Conclusion

In this assignment, we generated 2-D gaussian distributions for isotropic, diagonal and full covariance matrix. And we then also observed how the decision boundary is created between 2 gaussian classes for all three cases. The decision boundary is linear when covariances are equal and quadratic when they are unequal.